Fisher-Kolmogorov
Equation for neurodegenerative diseases

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1 Introduction

Neurodegenerative diseases like Alzheimer's, Parkinson's, and ALS involve progressive neuronal death caused by misfolded proteins that spread through brain tissue. These diseases follow a prion-like mechanism where pathogenic proteins convert normal proteins into toxic aggregates through conformational templating, creating cascading propagation across neural networks.

Mathematical modeling uses reaction-diffusion equations to capture this spreading process. The models treat protein aggregation as a nonlinear reaction coupled with anisotropic diffusion through brain tissue, reducing the complex biology to tractable equations governing protein concentration dynamics in space and time.

This project implements a physics-based reaction-diffusion model combining the Fisher-Kolmogorov equation with directional diffusion terms. The model simulates the growth kinetics and transport of specific misfolded protein. The unified modeling framework enables large-scale simulations of protein propagation across brain networks, revealing how local nucleation events evolve into widespread pathology. These models can predict disease progression patterns, identify vulnerable brain regions, and inform therapeutic strategies targeting early-stage protein aggregation before irreversible neuronal damage occurs.

2 Problem model definition

The problem in his mathematical form

$$\begin{cases} \dfrac{\partial c}{\partial t} - \nabla \cdot (D \cdot \nabla c) - \alpha c(1-c) = 0 & \text{in } \Omega, \\ \\ D \cdot \nabla c \cdot \vec{n} & \text{in } \partial\Omega, \\ \\ c(t=0) = c_0 & \text{in } \Omega. \\ \end{cases} \tag{2.1}$$

2.1 Kinetics of prion-like spreading

To describe the fundamental mechanism behind prion-like spreading, the paper gives us a simplified kinetic model, which distinguishes between two states of a protein: the natural healthy form p, and the misfolded, pathogenic form $\tilde{p}$. The misfolded proteins propagate through the brain by converting healthy proteins into their misfolded form, like a autocatalytic behaviour.
In the model are considered some kinetic processes:

- **Recruitment and conversion:** Healthy proteins bind to misfolded proteins and adopt the misfolded state at a rate k_{12} $p + \tilde{p} \xrightarrow{k_{12}} \tilde{p} + \tilde{p}$.
- **Fragmentation:** Misfolded proteins can fragment to generate new infectious seeds (even though we don't consider this in the final formulation).
- **Production and clearance:** Healthy proteins are continuously produced at a constant rate k_0 and cleared at rate k_1, while misfolded proteins are cleared at a separate rate $\tilde{k_1}$.
- **Diffusion:** Both types of proteins diffuse in the surrounding medium, characterized by diffusion coefficients D_p and $D_{\tilde{p}}$ respectively.

The model leads to a system of coupled PDEs governing the evolution of the concentrations of the two different types of proteins:

$$\frac{dp}{dt} = Div(D_p \nabla p) + k_0 - k_1 p - k_{12} p\tilde{p}$$

$$\frac{d\tilde{p}}{dt} = Div(D_{\tilde{p}}\nabla\tilde{p}) - \tilde{k_1}\tilde{p} + k_{12}p\tilde{p}$$

To simplify the system, the paper assumes that the concentration of healthy protein is significantly larger than that of the misfolded form (i.e., $p \gg \tilde{p}$), allowing a quasi-steady-state approximation for p. Under this assumption the healthy protein concentration is defined as:

$$p \approx \frac{k_0}{k_1 + k_{12}\tilde{p}}$$

And expanding this expression in a Taylor series around $\tilde{p} = 0$, it is substituted into the equation for , yielding a nonlinear reaction-diffusion equation.

To further simplify the model, it is introduced a normalized concentration for misfolded proteins, which is: $c = \frac{\tilde{p}}{\tilde{p_{max}}}$, where $0 \leq c \leq 1$.

This leads to a logistic-type equation:

$$\frac{\partial c}{\partial t} = \mathrm{Div}(D\nabla c) + \alpha c(1-c)$$

where $D = D_{\tilde{p}}$ is the diffusion tensor and α is an effective growth rate defined as:

$$\alpha = \frac{k_{12}k_0}{k_1} - \tilde{k}_1$$

This final equation captures the key dinamics of prion-like propagation: the diffusion of misfolded proteins and their growth due to conversion of healthy proteins. The growth rate α increases with the conversion rate k_{12} and the equilibrium healthy protein concentration $p_0 = k_0/k_1$, and decreases with the clearance rate of misfolded proteins $\tilde{k}_1$.

2.2 Fisher–Kolmogorov

To model the physics of prion-like diseases, we adopt the equation we have just seen, which is known in mathematical biology as the Fisher–Kolmogorov equation. Later on in this report we explain how to derive the weak formulation to solve this kind of equations with iterative algorithms.

2.3 Weak formulation

The following is the problem we were asked to solve:

$$\begin{cases} \dfrac{\partial c}{\partial t} - \nabla \cdot (D \cdot \nabla c) - \alpha c(1-c) = 0 & \text{in } \Omega, \\ \\ D \cdot \nabla c \cdot \vec{n} & \text{on } \partial\Omega, \\ \\ c(t=0) = c_0 & \text{in } \Omega. \\ \end{cases} \tag{2.2}$$

We developed the weak formulation ourselves, which is based on the following steps:

1. First we multiply everything times v, which is the value we have to find:
$$\int_\Omega \frac{\partial c}{\partial t} \cdot v - \int_\Omega \nabla \cdot (D \cdot \nabla c) \cdot v - \int_\Omega \alpha c(1-c) \cdot v dx = 0$$
for all $v \in V$, where V is the space of test functions H^1.

2. Then we integrate by parts the second term, which leads to the following expression:
$$\int_\Omega \frac{\partial c}{\partial t} \cdot v + \int_\Omega D \cdot \nabla c \cdot \nabla v - \int_\Omega \alpha c(1-c) \cdot v dx = 0$$

3. Then we use the backward Euler method to discretize the time derivative, which leads to the following expression:
$$\int_\Omega \frac{c_{n+1} - c_n}{\Delta t} \cdot v + \int_\Omega D \cdot \nabla c_{n+1} \cdot \nabla v - \int_\Omega \alpha c_{n+1}(1 - c_{n+1}) \cdot v dx = 0$$

4. Then, given the Frechet derivative:
$$lim_{\|d\| \to 0} \frac{|R(u+d) - R(u) - A(u)d|}{\|d\|} = 0$$

5. Now we can define the nonlinear residual:
$$\begin{aligned} R(c_{n+1} + d, v) = & \int_\Omega \frac{c_{n+1} + d - c_n}{\Delta t} v\, dx + \int_\Omega D \nabla (c_{n+1} + d) \cdot \nabla v\, dx \\ & - \int_\Omega \alpha (c_{n+1} + d)(1 - c_{n+1} + d)\, v\, dx = 0 \end{aligned}$$

6. Which simplifies to:
$$\begin{aligned} R(c_{n+1} + d, v) - R(c_{n+1}, v) = & \left(\int_\Omega \frac{c_{n+1} + d - c_n}{\Delta t} v\, dx - \int_\Omega \frac{c_{n+1} - c_n}{\Delta t} v\, dx \right) \\ & + \left(\int_\Omega D \nabla (c_{n+1} + d) \cdot \nabla v\, dx - \int_\Omega D \nabla c_{n+1} \cdot \nabla v\, dx \right) \\ & - \left(\int_\Omega \alpha (c_{n+1} + d)(1 - c_{n+1} - d) v\, dx - \int_\Omega \alpha c_{n+1}(1 - c_{n+1}) v\, dx \right) \end{aligned}$$

7. Finally, we make the following substraction:
$$\begin{aligned} R(c_{n+1}, v) - R(c_n, v) = & \int_\Omega \frac{c_{n+1} + d - c_n}{\Delta t} v\, dx + \int_\Omega D \nabla c_{n+1} \cdot \nabla v\, dx + \int_\Omega D \nabla d \cdot \nabla v\, dx \\ & - \int_\Omega \alpha \left(c_{n+1} + d - c_{n+1}^2 - 2d c_{n+1} - d^2 \right) v\, dx \\ & - \int_\Omega \frac{c_{n+1} - c_{n+1}}{\Delta t} v\, dx - \int_\Omega D \nabla c_{n+1} \cdot \nabla v\, dx \\ & + \int_\Omega \alpha c_{n+1}(1 - c_{n+1})\, v\, dx \end{aligned}$$

8. We can approximate $(d^k)^2$ as a higher-order small term, leading to the following expression:

$$R(c_{n+1} + d^k, v) - R(c_{n+1}, v) = \int_\Omega \frac{d^k}{\Delta t} v \, dx + \int_\Omega D \nabla d^k \cdot \nabla v \, dx$$
$$- \int_\Omega \alpha d^k (1 - 2c^k_{n+1}) \, v \, dx + \mathcal{O}\left((d^k)^2\right)$$

9. The right-hand side corresponds to the action of the Fréchet derivative:

$$\frac{dR}{dc} \cdot c_{n+1}(d, v) = a(c_{n+1})(d, v)$$

2.4 Space discretization

2.5 Time discretization

Guided by the original authors, the time discretization approach centers on the backward Euler iterative method, which provides the stability necessary for handling the nonlinear reaction-diffusion dynamics inherent in prion spreading models regardless of the Δt used.
At each iteration, we compute an approximation of the concentration $c(t)$ and its derivative with respect to time using the formula $\frac{\delta c}{\delta t} = \frac{c_n + 1 - c_n}{\Delta t}$. This choice is particularly significant because it transforms our original time-dependent parabolic problem into a sequence of quasistatic, elliptic-type problems. In each time step, we solve for the unknown c_{n+1} using the known solution from the previous step c_n. The implicit nature of this scheme means we are always solving "forward in time" but using information from the future time level, which drastically improves stability.

2.6 Finate element setting and final assembly

3 Set up and computation

3.1 Parameters

3.1.1 The mesh

3.2 1D and 3D: Key aspects

3.2.1 Similarity

3.2.2 Differences

3.2.3 Parallelization

4 Performance

4.1 Different parameters

4.2 Convergence

5 Final results